Vadla Pranav

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GitHub | LinkedIn

Professional Summary

AI/ML Engineering Intern experienced in logistics optimization, cybersecurity, and intelligent application development. Skilled in Python, machine learning, mathematical optimization, backend APIs, and database-driven systems. Experienced in integrating optimization engines with existing applications and developing data-driven solutions for real-world operational problems.

Technical Skills

Languages: Python, Java, C++, C, SQL, JavaScript

AI/ML: Scikit-learn, TensorFlow, Keras, CNNs, Transformers, Feature Engineering, Classification, Model Evaluation

Optimization: Google OR-Tools, CVRP, Route Optimization, Cost Optimization, Vehicle Assignment, Operations Research

Backend & APIs: Node.js, Express.js, FastAPI, Flask, REST APIs, Python Subprocess Integration

Databases & Tools: MongoDB, SQLite, SQLAlchemy, Git, GitHub, Linux, Ruff, OSRM Routing

Web Technologies: Angular, HTML, CSS

Experience

AI/ML Intern – Mahindra Logistics Ltd *Jun 2026 – Present*

- Developing logistics planning and optimization workflows using Python, MongoDB, and Google OR-Tools, with emphasis on vehicle feasibility, utilization, routing, and pricing.
- Implementing cost- and turnaround-time (TAT)-optimized planning with vendor and vehicle alternatives, order recovery, and explanations for unsuccessful orders.
- Integrating Python optimization services with Angular and Express through FastAPI, REST APIs, MongoDB-backed caches, job execution, polling, and rerun controls.

Research & Development Intern – CyberGuard360 (Remote) *Jul 2024 – Dec 2025*

- Developed an LLM-assisted phishing simulation platform with backend services for campaign execution, analytics, and reporting.
- Built Cyber Threat Intelligence pipelines for URL and IP reputation analysis and developed hybrid machine learning and rule-based approaches for malicious URL detection.
- Integrated sandbox-based behavioral analysis into cybersecurity detection workflows and applied feature engineering to improve classification.

Projects

Cost, TAT & Backhaul Optimization System *2026*

- Developed cost- and TAT-optimization workflows covering order grouping, vehicle assignment, routing, commercial pricing, alternative recommendations, and failed-order reporting.
- Consolidated planner execution into four phases and integrated optimization services with Angular, Express, FastAPI, and MongoDB for cached results and controlled reruns.
- Integrated a separately developed backhaul engine into the planner workflow, including recommendation filtering, vehicle mapping, availability checks, and reservation safeguards.
- In a synthetic test, consolidated 36 orders into one feasible trip and reduced estimated cost from Rs. 19,696.16 to Rs. 4,919.20 (75.02%). The TAT optimizer produced a feasible plan with an 18.39-hour maximum TAT.

Phishing Simulation Platform Using LLMs *2025*

- Built an LLM-assisted phishing simulation platform for cybersecurity awareness, with campaign execution and reporting.
- **Technologies:** Python, Flask, SQLite, REST APIs, LLMs

URL Threat Classifier *2025*

- Developed a malicious URL classification system using static and dynamic features with hybrid machine learning techniques; reported over 90% detection accuracy.
- **Technologies:** Python, Scikit-learn, Flask, Feature Engineering

Self-Driving Solar Car *2025*

- Developed a solar-powered autonomous RC car using CNN-based lane detection, computer vision, and obstacle avoidance.

- **Technologies:** Python, TensorFlow, CNN, Raspberry Pi, Computer Vision

Education

B.Tech in Computer Science and Engineering *2023 – 2027*
Keshav Memorial Engineering College (KMEC), Hyderabad – CGPA: 8.54

Intermediate (MPC) – *2023*
Narayana Junior College – Percentage: 96%

SSC – *2021*
St. Paul's High School – CGPA: 10.0

Achievements

- 2nd Runner-Up, MedFusion Hackfest 2026 (Team Lead).
- Runner-Up, TECHTONIC'24 Autonomous Systems Competition.